

PATARA TRIRAT

AI Research Scientist

KAIST Ph.D.

Current Affiliation: Principal AI Research Scientist at DeepAuto.ai

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— RESEARCH INTERESTS

As an AI research scientist, I am passionate about advancing *agent-based task automation* toward human-agent collaboration and exploring new frontiers of perception in *autonomous agents operating in complex, real-world environments* through multimodal large language models. My research interests include

Agentic AI & LLM Agents: TS-Debate [[arXiv](#)], AutoML-Agent [[Paper](#)], Agentic Predictor [[Paper](#)], MONAQ [[Paper](#)], ZeroPOIRec [[Paper](#)], OmniRetrieval [[arXiv](#)]

Spatiotemporal Data Mining: DF-TAR [[Paper](#)], MG-TAR [[Paper](#)], Universal Time Series Survey [[Paper](#)], Time-PEFT [[Paper](#)], MacroLens [[arXiv](#)]

AutoML: AutoML-Agent [[Paper](#)], Conversational AutoML [[Paper](#)], PASTA [[Paper](#)], TFAS [[Paper](#)]

— EDUCATION

AUGUST 2020 – FEBRUARY 2025 / DAEJEON, SOUTH KOREA

KAIST / Ph.D in School of Computing

Dissertation: Neural Architecture Search for Time-Series Analysis with Context-Specific Performance Estimators

Advisor: Prof. Jae-Gil Lee

AUGUST 2018 – AUGUST 2020 / DAEJEON, SOUTH KOREA

KAIST / M.S. in Knowledge Service Engineering (Currently, Data Science)

Thesis: Analysis of Dangerous Driving Behavior from Driving Log Data for Traffic Accident Risk Prediction

Advisor: Prof. Jae-Gil Lee

JUNE 2012 – MAY 2016 / BANGKOK, THAILAND

Kasetsart University / B.S. in Computer Science with First Class Honors

Advisor: Prof. Pakaket Wattuya

— EMPLOYMENT

JANUARY 2025 — PRESENT / SEOUL, SOUTH KOREA

Principal AI Research Scientist at **DeepAuto Co., Ltd.**

Multi-Agent System Development, Automated Design of Multimodal Agentic AI Systems

JANUARY 2024 — MAY 2024 / SEOUL, SOUTH KOREA

AI Research Intern at **DeepAuto Co., Ltd.**

Large Language Model for AutoML using Multi-Agent Framework

MAY 2017 — AUGUST 2018 / BANGKOK, THAILAND

Frontend Developer at **Sobtid Edutainment Co., Ltd.**

APIs and Web Applications of Exam and Test Preparation Systems for Clients and In-House Officers

— SELECTED PUBLICATIONS

Conference Proceedings

[C10] [Trirat, P.](#), Kwak J.M., Heo J., Lee H., and Hwang, S.J. *Multimodal Collaborative Debate for Zero-Shot Time Series Reasoning*

Conference on Empirical Methods in Natural Language Processing (EMNLP) 2026

[C9] Na, J., [Trirat, P.](#), Park, C., and Lee, J. *Time-PEFT: Temporal and Multichannel Complexity-Based Fine-Tuning for Time-Series Foundation Models*

International Conference on Machine Learning (ICML) 2026

[C8] [Trirat, P.](#), Jeong, W., and Hwang, S.J. *Multi-View Encoders for Performance Prediction in LLM-Based Agentic Workflows*

Presented at ICML 2025 Workshop on Multi-Agent Systems (**Oral Presentation, Outstanding Paper**)

International Conference on Learning Representations (ICLR) 2026

[C7] [Trirat, P.](#) and Lee, J. *MONAQ: Multi-Objective Neural Architecture Querying for Time-Series Analysis on Resource-Constrained Devices*

Presented at ICML 2025 Workshop on Multi-Agent Systems

Findings of the Association for Computational Linguistics: EMNLP 2025

[C6] [Trirat, P.](#), Jeong, W., and Hwang, S.J. *AutoML-Agent: A Multi-Agent LLM Framework for Full-Pipeline AutoML*

International Conference on Machine Learning (ICML) 2025

[C5] Nam, Y., Trirat, P., Kim, T., Lee, Y., and Lee, J. *Context-Aware Deep Time-Series Decomposition for Anomaly Detection in Businesses* (**Oral Presentation**)
European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML PKDD) 2023
 [C4] Trirat, P.*, Nam, Y.*, Kim, T., and Lee, J. *AnoViz: A Visual Inspection Tool of Anomalies in Multivariate Time Series* (**Demonstration Track**)
AAAI Conference on Artificial Intelligence (AAAI) 2023
 [C3] Trirat, P. and Lee, J. *DF-TAR: A Deep Fusion Network for Citywide Traffic Accident Risk Prediction with Dangerous Driving Behavior* (**Oral Presentation**)
The Web Conference (TheWebConf) 2021
 [C2] Trirat, P., Noree, S., and Yi, M.Y. *IntelliMOOC: Intelligent Online Learning Framework for MOOC Platforms*
International Conference on Educational Data Mining (EDM) 2020
 [C1] Shin, Y., Yoon, S., Trirat, P., and Lee, J. *CEP-Wizard: Automatic Deployment of Distributed Complex Event Processing* (**Demonstration Track**)
IEEE International Conference on Data Engineering (ICDE) 2019

Journal Articles

[J5] Trirat, P., Shin, Y., Kang, J., Nam, Y., Na, J., Bae, M., Kim, J., Kim, B., and Lee, J. *Universal Time-Series Representation Learning: A Survey*
ACM Computing Surveys (SCIE, Impact Factor: 30.4)
 [J4] Kim, J., Seo, Y., Kim, Y., Kang, J., Shin, J., Trirat, P., and Lee, J. *Large Language Models are Zero-Shot Point-of-Interest Recommenders*
 Presented at **ECML PKDD 2025 Journal Track** (**Oral Presentation**)
Data Mining and Knowledge Discovery (SCIE, Impact Factor: 5.5)
 [J3] Trirat, P. and Lee, J. *TFAS: Zero-Shot NAS for General Time-Series Analysis with Time-Frequency Aware Scoring*
 Presented at **ECML PKDD 2025 Journal Track** (**Oral Presentation**)
Machine Learning Journal (SCIE, Impact Factor: 4.9)
 [J2] Trirat, P. and Lee, J. *PASTA: Neural Architecture Search for Anomaly Detection in Multivariate Time Series*
IEEE Transactions on Emerging Topics in Computational Intelligence Vol. 9 No. 4 (SCIE, Impact Factor: 6.5)
 [J1] Trirat, P., Yoon, S., and Lee, J. *MG-TAR: Multi-view Graph Convolutional Networks for Traffic Accident Risk Prediction*
IEEE Transactions on Intelligent Transportation Systems Vol. 24 No. 4 (SCIE, Impact Factor: 9.1)

Preprint / Ongoing Work

[O5] Trirat, P., Ha, J., Joo, H., Heo J., Lee H., and Hwang, SJ. *Proactive Problem Solver: LLM Agents for the Enterprise R&D Pipeline*
Ongoing Work
 [O4] Trirat, P., Heo J., Lee H., and Hwang, SJ. *Visual Anchoring for Scenario-Guided Forecasting*
Ongoing Work
 [O3] Trirat, P., Kwak J.M., Heo J., Lee H., and Hwang, SJ. *MacroLens: A Multi-Task Benchmark for Contextual Financial Reasoning under Macroeconomic Scenarios*
<https://arxiv.org/abs/2606.24950>
 [O2] Trirat, P., Zhang, S., Heo J., and Hwang, SJ. *Agentic Tool Generation with Hierarchical Goal-Driven Co-Planning*
Ongoing Work
 [O1] Baek, J., Jeong, S., Park, S., Yeo, W., Kang, M., Trirat, P., Lee, H., Hwang, SJ. *OmniRetrieval: Unified Retrieval across Heterogeneous Knowledge Sources*
<https://arxiv.org/abs/2605.29250>

— SERVICES

Conference Reviewer: KDD, SDM, ICML, ICLR, NeurIPS, AAAI, ICML MAS, ICLR LLA
Journal Reviewer: TMLR, ACM CSUR, IEEE T-ITS, IEEE TNNLS, IEEE TMC, IEEE TETCI
Research Mentoring: Mr. Phatarapran Saraluck (Undergraduate), Mr. Wirittipol Supdateyarnnakorn (Undergraduate), Mr. Geonju Lee (Research Intern / Master's Student), Mr. Jisung Ha (Research Intern)
Additional Activity: KAIST Student Ambassador, KAIST ISSS Student Assistant Photographer

— LANGUAGES AND SKILLS

Languages: **Thai:** Native | **English:** Effective Operational Proficiency (TOEIC: 975 / 990) | **Korean:** Effective Operational Proficiency (TOPIK IBT: 6 / 6)
Programming Languages: Python, JavaScript, HTML/CSS, R
Frameworks & Libraries: Tensorflow, Keras, PyTorch, Agents SDK, LangChain/LangGraph, React.js, Node.js

— ACCOMPLISHMENTS

Honors and Awards

- + 2026 **Best Reviewers** (Research Track) by KDD 2026
- + 2026 **2nd Place** of XForecast Challenge at KDD 2026 Finance Day by LG AI Research
- + 2026 **Gold Reviewer Award** (Top 25%) by ICML 2026
- + 2025 **Outstanding Paper Award** by ICML 2025 Workshop on Multi-Agent Systems in the Era of Foundation Models: Opportunities, Challenges and Futures
- + 2025 **Commendable Reviewer** by KDD (February Cycle) 2025
- + 2025 **Top 10% Outstanding Reviewer** by KDD (August Cycle) 2025
- + 2023-24 **Student Ambassador** by KAIST Office of Admissions
- + 2023 University Visit Program Scholarship Recipient by KAIST School of Computing
- + 2022 외국인 유학생 독도사랑 한국어 말하기대회 최우수상 (팀명: K-GOOSE) by Daegu University
- + 2018-24 KAIST Scholarship (Master and Doctoral Program) by KAIST
- + 2014-16 **Faculty of Science Awards** for Outstanding Academic Achievement Student by Faculty of Science, Kasetsart University

Invited Talks

- + 2021 International Seminar on Policies for Older Drivers
- + KSC2021 Top Conference Session